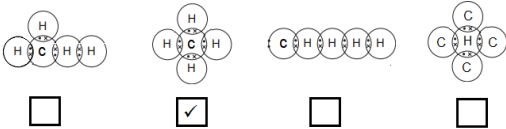


| Question  |                    |                              |                    | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks available |                    |                         |                   |       |      |                           |               |   |      |                              |                    |   |      |                              |                  |  |   |  |   |  |  |
|-----------|--------------------|------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------|-------------------------|-------------------|-------|------|---------------------------|---------------|---|------|------------------------------|--------------------|---|------|------------------------------|------------------|--|---|--|---|--|--|
|           |                    |                              |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                       | AO1             | AO2                | AO3                     | Total             | Maths | Prac |                           |               |   |      |                              |                    |   |      |                              |                  |  |   |  |   |  |  |
| 3         | (a)                | (i)                          | I                  | <div><br/><input type="checkbox"/> <input checked="" type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/></div> <p>accept any correct indication of the answer</p>                                                                                                                                                                                    | 1               |                    |                         | 1                 |       |      |                           |               |   |      |                              |                    |   |      |                              |                  |  |   |  |   |  |  |
|           |                    |                              | II                 | <div><span>covalent</span>    ionic    giant covalent    metallic</div> <p>accept any correct indication of the answer</p>                                                                                                                                                                                                                                                                                                                            | 1               |                    |                         | 1                 |       |      |                           |               |   |      |                              |                    |   |      |                              |                  |  |   |  |   |  |  |
|           |                    | (ii)                         | I                  | <table><tr><th>Substance</th><th>Melting point / °C</th><th>Electrical conductivity</th><th>Type of structure</th></tr><tr><td>A</td><td>2072</td><td>conducts only when molten</td><td>(giant) ionic</td></tr><tr><td>B</td><td>-182</td><td>does not conduct electricity</td><td>(simple) molecular</td></tr><tr><td>C</td><td>1610</td><td>does not conduct electricity</td><td>(giant) covalent</td></tr></table> <p>all <b>three</b> correct</p> | Substance       | Melting point / °C | Electrical conductivity | Type of structure | A     | 2072 | conducts only when molten | (giant) ionic | B | -182 | does not conduct electricity | (simple) molecular | C | 1610 | does not conduct electricity | (giant) covalent |  | 1 |  | 1 |  |  |
| Substance | Melting point / °C | Electrical conductivity      | Type of structure  |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |                    |                         |                   |       |      |                           |               |   |      |                              |                    |   |      |                              |                  |  |   |  |   |  |  |
| A         | 2072               | conducts only when molten    | (giant) ionic      |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |                    |                         |                   |       |      |                           |               |   |      |                              |                    |   |      |                              |                  |  |   |  |   |  |  |
| B         | -182               | does not conduct electricity | (simple) molecular |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |                    |                         |                   |       |      |                           |               |   |      |                              |                    |   |      |                              |                  |  |   |  |   |  |  |
| C         | 1610               | does not conduct electricity | (giant) covalent   |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |                    |                         |                   |       |      |                           |               |   |      |                              |                    |   |      |                              |                  |  |   |  |   |  |  |
|           |                    |                              | II                 | substance <b>B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 | 1                  |                         | 1                 |       |      |                           |               |   |      |                              |                    |   |      |                              |                  |  |   |  |   |  |  |
|           | (b)                |                              |                    | 14.28 / 14.3 / 14      award (2) for correct answer<br><br>award (1) for correct molecular mass of 56 if incorrect answer<br><br>no ECF if incorrect $M_r$ calculated in first step                                                                                                                                                                                                                                                                   |                 | 2                  |                         | 2                 | 2     |      |                           |               |   |      |                              |                    |   |      |                              |                  |  |   |  |   |  |  |
|           |                    |                              |                    | Question 3 total                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2               | 4                  | 0                       | 6                 | 2     | 0    |                           |               |   |      |                              |                    |   |      |                              |                  |  |   |  |   |  |  |